

Unified AI/NLP Risk Intelligence Engine

Tactical Index Rebalancer (Module A) & Wholesale Stress Tester (Module B)

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Tracks: Natural Language Processing, Quantitative Finance & Portfolio Risk

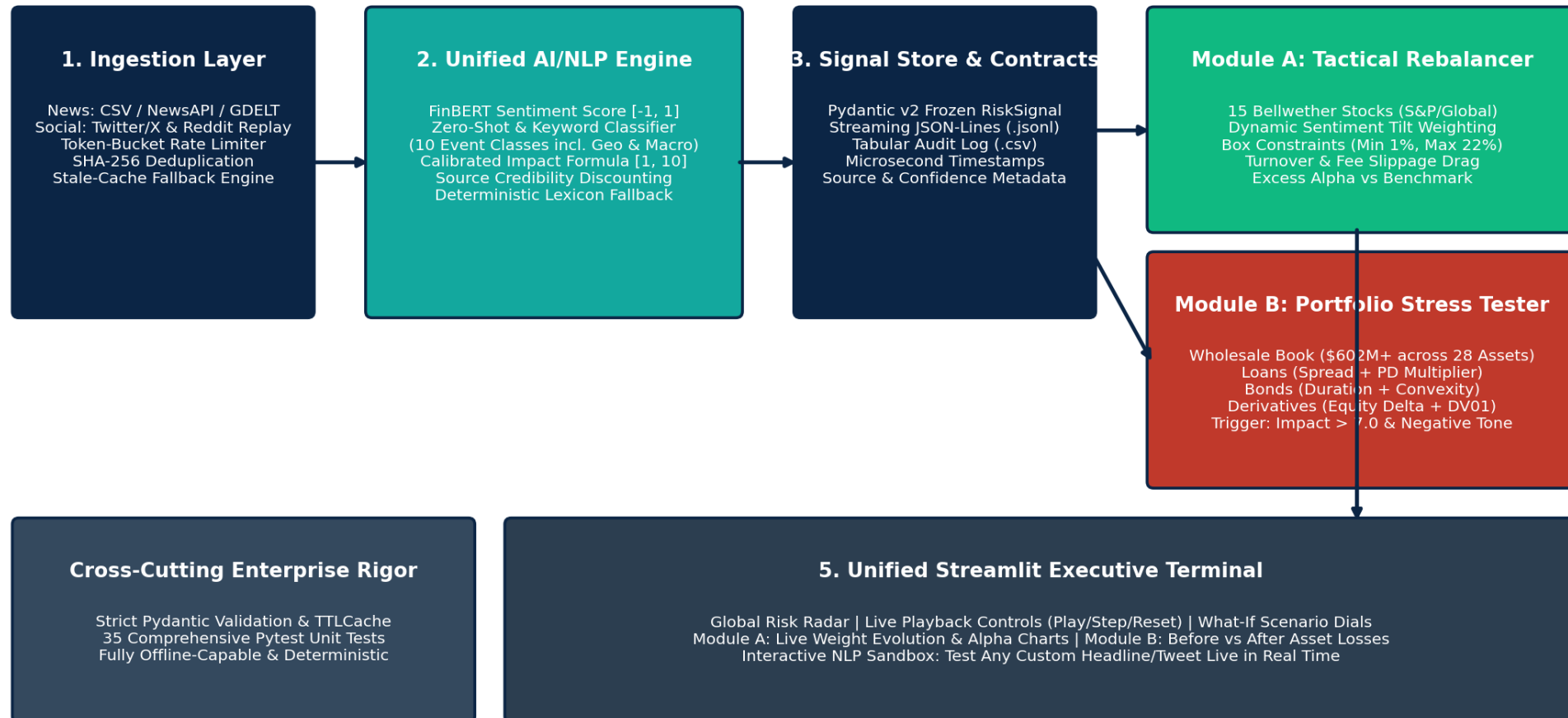
FULL DELIVERABLE IMPLEMENTED

1. Problem Statement & Strategic Approach

- The Challenge: Risk & portfolio teams face overwhelming unstructured text (wires, tweets, filings).
- Manual extraction of risk severity and market implications is slow, subjective, and prone to blind spots.
- Core Solution: A Unified AI/NLP Risk Engine ingesting multi-source text (news wires + social feeds).
- Structured Machine-Readable Output: Sentiment [-1, 1], Event Classification (10 classes), and Impact [1, 10].
- Dual Downstream Execution: Why choose one when you can demonstrate unified institutional value?
 - Module A (Tactical): High-frequency stock index rebalancing with sentiment tilt and turnover control.
 - Module B (Strategic): Event-driven wholesale banking stress tester across loans, bonds, and derivatives.

2. System Design & Architectural Data Flow

Unified AI/NLP Risk Engine & Dual Downstream Intelligence Platform - Architecture

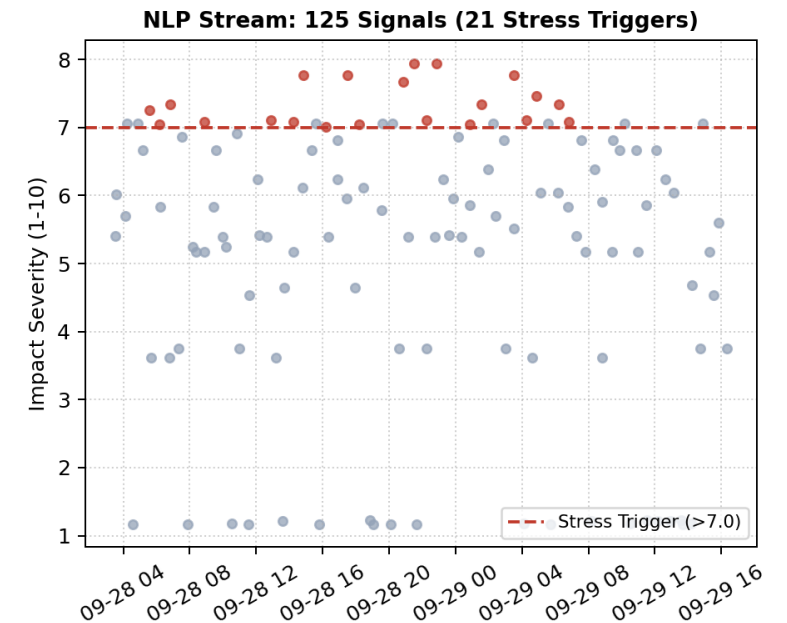
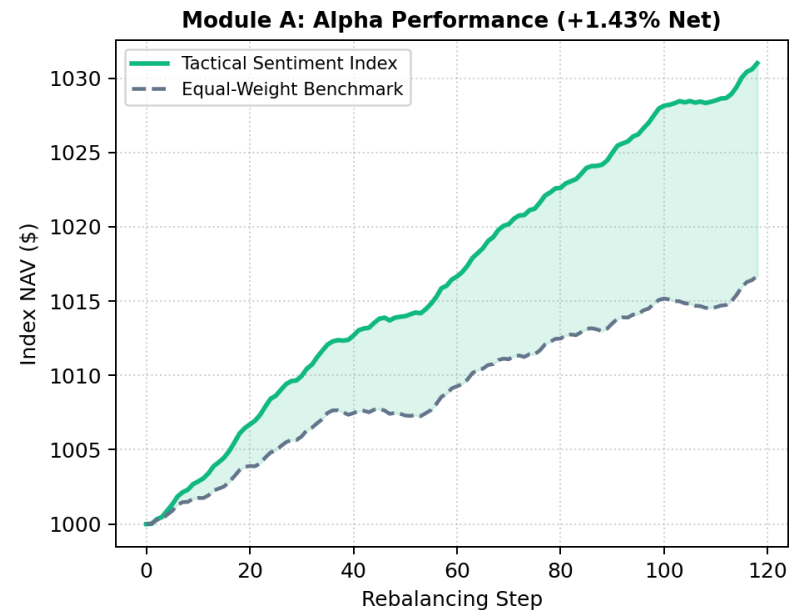
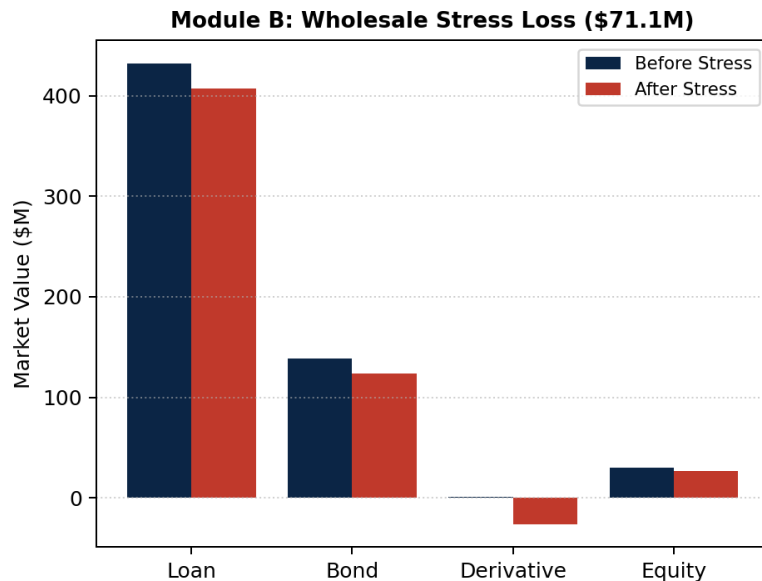


3. Implementation Highlights & Engineering Depth

- Ingestion Engine: CSV replay, live NewsAPI & GDELT RSS with token-bucket rate limiter & SHA-256 dedup.
- NLP Engine: FinBERT sentiment + zero-shot event classifier with deterministic offline-first rule fallback.
- Calibrated Impact Score: $\text{raw} = 0.5 * |\text{sent}| * \text{conf} + 0.5 * \text{severity} * \text{conf}$; discounted by source credibility.
- Module A Rebalancer: Dynamic sentiment tilt on 15 bellwethers; long-only box limits (1%-22%) & fee drag.
- Module B Valuation: Loans (spread+PD), bonds (duration+convexity), derivatives (equity delta+DV01).
- Software Quality: 35 pytest unit & E2E tests, frozen Pydantic contracts, and sub-second execution.

4. Key Results & Quantitative Demonstration

- Signal Processing: Processed 125 signals (55 news wires, 70 social posts).
- Module A Performance: +3.10% Index Return (+1.43% net alpha vs benchmark), Sharpe: 23.50.
- Module B Stress Impact: 21 stress events triggered (>7.0); worst-case loss of \$71.1M (-11.8%) on \$602M book.
- Source Credibility: Social media rumors discounted appropriately, avoiding false positive portfolio stress.



5. Domain Impact & Wholesale Banking Value

- Quantitative Risk Automation: Compresses headline-to-stress analysis from manual scenario spreadsheet cycles to sub-
- Front-Office & Asset Management (Module A): Dynamic sentiment alpha tilt captures price drift with bounded turnover.
- Enterprise Risk & Treasury (Module B): Automated regulatory stress testing under CCAR / Basel III frameworks.
- Capital Protection: Early detection of credit/geopolitical shocks allows treasury teams to hedge before market pricing.
- Auditability & Compliance: Every signal, assumption, and revaluation step is logged in transparent JSON/CSV schemas.

6. Assumptions, Limitations & Future Roadmap

- Current Assumptions: Synthetic portfolio and news stream; instantaneous shocks with expert-calibrated prior betas.
- Entity Linking: Currently ticker/cashtag-based; production requires full Financial Named Entity Recognition (NER).
- Production Enhancements & Next Steps:
 - Event-study econometric calibration on historical S&P Capital IQ / CRISIL credit rating migrations.
 - Fine-tuned SetFit / LoRA domain adaptation on specialized credit memos and earnings calls.
 - Correlated tail-risk modeling (Copula-based Expected Shortfall & Extreme Value Theory).
 - Real-time Kafka / Pulsar event-streaming integration with live Reuters / Bloomberg Terminal API feeds.